

A	B	C	F() Boolean equation	AND/OR circuit	NAND circuit
0	0	0	0	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	1	1	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	1	1	1

Using De Morgan's theorem and NAND gate equivalents for AND, OR, and NOT, the NAND-based version replicates the logic function entirely with NAND gates.

By simulating both circuits in Logisim evolution, and filling in the truth table, we confirm that for every input combination, both circuits output the same result.

This proves that the NAND circuit is functionally equivalent.